

2025

无人渡我，我亦成舟

我最大的依靠不是别人，而是自己；面对人生的风浪，我既是那个需要过河的人，也是那艘唯一能载我过河的船。

January & February, 2025

Weeks

Interests

- 1. ✓ 撰写个人 2024 年度总结报告;
- 2. ✓ 起笔大学四年总结报告;
- 3. ✓ 构建起基本完整的社交网络，以及关系的定量分析;
- 4. ✗ 构建西电计科传承指南站点;

Learning

- 1. ✗ 完成 GAMES101 课程学习以及图形学领域的相关知识;

Research

- 1. ✗ 总结鲁棒性提示学习的研究工作，汇总相关思路笔迹，归档科研习惯、写作技巧等心得;
- 2. ✓ 阅读三维人脸重建领域的研究论文，复现最新研究成果以及探究科研思路;

March, 2025

Week 1: 了解研究方向

February 14

- 1. ✓ Read *Gaussian Head Avatar: Ultra High-fidelity Head Avatar via Dynamic Gaussians*;
- 2. ✓ Read 《三维人脸成像及重建技术综述》and summary;

February 18

- 1. ✓ Set up data collection and processing pipeline—VHAP;

February 19

- 1. ✓ Read *Learning a model of facial shape and expression from 4D scans* and make a summary

February 20

- 1. ✓ Finished configuring the experimental environments and training pipelines—GaussianAvatar;

Week 2: 完成毕设中期工作

研究 GaussianAvatars 论文及项目，完成毕设中期检查报告

February 21

- 1. ✓ Read *GaussianAvatars: Photorealistic Head Avatars with Rigged 3D Gaussians* and make a summary;

February 23

- 1. ✓ Explore the VHAP working pipelines and learn the FLAME model;

Week 3: 调研数字人 Audio-Driven 研究脉络

- 1. 调研方法和目标
 - 1. 搜集近年来相关研究论文，整理归纳;
 - 2. 阅读每一篇论文并进行简要总结：总结研究任务，方法，存在的挑战;
 - 3. 汇总所有论文总结，形成综述报告;
- 2. 论文追溯
 - 1. Inspiration: 从论文中追溯启发的文章;
 - 2. Basic Model: 从论文中积累经典的模型结构;

March 4

- 1. ✓ Read EMO: Emote Portrait Alive - Generating Expressive Portrait Videos with Audio2Video Diffusion Model under Weak Conditions and make a summary;

March 5

- 1. ✓ Read MO2: End-Effector Guided Audio-Driven Avatar Video

Generation and make a summary;

March 6

- 1. ✓ Read VividTalk: One-Shot Audio-Driven Talking Head Generation Based on 3D Hybrid Prior and make a summary;
- 2. ✓ Read Lip Movements Generation at a Glance and make a summary;

March 7

- 1. ✓ Read [GAN-Based] [CVPR 2017 Speech2Vid] You said that? and make a summary;
- 2. ✓ Read [GAN-Based] [CVPR 2019 FacialAnimation] Realistic Speech-Driven Facial Animation with GANs and make a summary;

Week 4: Paper Reading

March 10

- 1. ✓ Determine the roadmap of research survey;

March 18

- 1. ✓ Determine the research direction of 3D Real-time Audio-driven Talking Head Generation;

March 23: Adjustment

- 1. ✓ Reading paper in the morning and coding in the afternoon;
- 2. ✓ Learning in the evening and make a daily summary;

Week 5: Paper Reading

March 24

- 1. ✓ Read AD-NeRF, ER-NeRF and make a summary;
- 2. ✓ Read the codes of GuassianTalker;

March 25

- 1. ✓ Read the Efficient Region-Aware Neural Radiance Fields for High-Fidelity Talking Portrait Synthesis and make a summary;

March 26

- 1. ✓ Read the Real-time Neural Radiance Talking Portrait Synthesis via Audio-spatial Decomposition and make a summary;
- 2. ✓ Summarize the section 1 of Scratchapixel;

March 31

- 1. ✓ Add the UNet Network to enhance the GaussianAvatar and finish coding;
- 2. ✓ Finish writing the Introduction of the undergraduate thesis;

April, 2025

- 1. Adjust Shedules
 - 1. Spend two weeks on finishing the research survey and the summary slides;
 - 2. The last two weeks will be taken to finish the undergraduate thesis.
- 2. Fruit Display
 - 1. A good designed and resealed slide;
 - 2. A well organized thesis and the prepared project;

Week 1: Research Survey and Not Continued

April 1

- 1. ✓ Scan the collected papers;
- 2. ✓ Make the template of summarized slides;

April 2

- 1. ✓ Read a research survey and make a summary;

Week 2: Quickly Overcame the Undergraduate Thesis

- 1. ✓ Finished the undergraduate thesis;

Week 3 to 4: I don't know. It's a mess.

- 1. ✗ What I have did? I don't know.
- 2. ✓ Read the LAM paper and tried to reproduce it.
- 3. ✓ It seemed that I was also dedicated to 3DAvatar Viewer.

May, 2025

Week 1: Labour Vacation

May 1

- 1. ✓ Have a good rest in my room and have nothing to do;

May 2

- 1. ✓ Go to the laboratory and have a short walk along the Shenzhen Bay at dusk;

May 3

- 1. ✓ Have a meal with relatives and friends;

May 4

- 1. ✓ Go to the Lotus Pond Park with relatives and friends;

Week 2: Debugging the 3D Avatar Viewer

May 5

- 1. ✓ Summary all things and stay motivated;
- 2. ✓ Have wrote an essay about my trip in Labour Holiday;
- 3. ✓ Debugging the 3D Avatar Viewer and seemed have founded one problem.

May 6

- 1. ✓ Working: Retraining the GaussianAvatars with saving global position, scale, rotation;
- 2. ✓ Working: Failed debugging and change to reproduce the OpenChatAvatar;
- 3. ✓ Wealther: Dense rain, thick fog and rolling thunder;

May 7

- 1. ✓ Working: Have a better understanding and verify the data loader, but not be sure yet;
- 2. ✓ Wealther: Dense rain in the morning and become wet in the road;

May 8

- 1. ✓ Working: Have founded a good problem in 3DAvatar Viewer but still struggled to solve it;
- 2. ✓ Emotion: Tired of this debugging and go home directly at night;

May 9

- 1. ✓ Working: Still keep motivated to face with it and I don't know what will happen; —writing in the morning
- 2. ✓ Plan: Just to decompose the long JavaScript codes into several well-organized files and make a summary about the codes structure;
- 3. ✓ Emotion: A little excited because of the Friday;

May 10

- 1. ✓ Vocation: Have a good Saturday and reset without going out.

May 11

- 1. ✓ Doing: A prepared day for returning to Xian.
- 2. ✓ Emotion: Have a dinner together with all senior brothers.

Week 3: 本科优秀毕业生评选

- 1. Meeting
 - 1. Have a talk and a photo together with the reachable friends;
 - 2. Go to dinner with all roommates and pull the hearts together.
- 2. Working
 - 1. Update the blog of SRRS and the github repository.
 - 2. Advance the undergraduate thesis and make sure all materials best and ready;
 - 3. Struggle to finish the 3D Avatar Viewer. Must be successful
- 3. Hobbies
 - 1. Release a blog about the logging media and summary sheet;
 - 2. Open a gallery page in blog to show the whole life trip;
 - 3. Go to have a good look about the Xidian University and take photos for essay writing.
- 4. Living
 - 1. Sell the unused things to the next generation for clean up old but still good things.

May 12

- 1. ✓ Trip: Taking a flight from Shenzhen to Xian.
- 2. ✓ Shedules: Making several working and living tasks to finish.

ish.

May 13

- 1. ✗ Trip: Have forgot what I did.

May 14

- 1. ✓ Working: Reproduce the results of the 3D Gaussian Avatar Viewer.

May 15

- 1. ✓ Hobbies: Finishe the goal of Open a gallery page.

May 16

- 1. ✓ Meeting: Have a hearty dinner with some roommates.

May 17

- 1. ✓ Working: Testing the render but got bad results.

Week 4: 本科毕设之软硬件验收

May 19

- 1. ✓ Working: Finding the problem of GaussianAvatar Viewer and achieved the static rendering.
- 2. ✓ Meeting: Have a good dinner and rest with best friends.

May 20

- 1. ✓ Working: Just do nothing but tuning the LAM viewer without any production.
- 2. ✓ Fitness: Running for 2km in the playground but got hurt on foot.

May 21

- 1. ✓ Working: Make sure to be fully prepared for tomorrow's hardware and software inspection.
- 2. ✓ Rest: Just have a good rest for my hurt foot.

May 22

- 1. ✓ Working: Successfully passed hardware and software inspection and got the high score of 94.

May 23

- 1. ✓ Living: Still have a day spent in the dorm.
- 2. ✓ Working: Revised my undergraduate thesis and then finalized it.

May 24

- 1. ✓ Working: Upload my undergraduate thesis and preparing the presentation slides for the final defense.

Week 5: 本科毕设之终期答辩

May 27

- 1. ✓ Working: Apply the honor of graduate star, preprint the thesis and materials, make the slides.

May 28

- 1. ✓ Working: Attend to the final defense and got the highest score in the group.

May 29

- 1. ✓ Working: Modified my undergraduate thesis according to the suggestions of reviewers.

May 30

- 1. ✓ Meeting: Get the graduation project documents stamped.

May 31

- 1. ✓ Meeting: Finish the goal of having a big dinner with all roommates.

June, 2025

Week 1

June 1

- 1. ✓ Working: Get all materials of graduated thesis prepared to upload.

July, 2025

Week 1
Header 2

August, 2025

Week 1
Header 2

September, 2025

Week 1
Header 2

October, 2025

Week 1
Header 2

November, 2025

Week 1
Header 2

December, 2025

Week 1
Header 2